

TRIGGERS & PROCEDURES

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1. CREATE DATABASE

The screenshot shows the SQL Developer interface with a query window containing two commands:

```
1 • CREATE DATABASE BankDB;  
2 • USE BankDB;
```

The Output window displays the execution results:

#	Time	Action	Message
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	10:17:58	USE BankDB	0 row(s) affected

2. CREATE CUSTOMER TABLE

The screenshot shows the SQL Developer interface with a query window containing three commands:

```
1 • CREATE DATABASE BankDB;  
2 • USE BankDB;  
3 • CREATE TABLE Customers (  
4     Customer_ID INT PRIMARY KEY,  
5     Customer_Name VARCHAR(100) NOT NULL,  
6     Phone VARCHAR(15),  
7     Email VARCHAR(100),  
8     City VARCHAR(50)  
9 );  
10  
11
```

The Output window displays the execution results:

#	Time	Action	Message
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	10:17:58	USE BankDB	0 row(s) affected
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected

3. CREATE ACCOUNT TABLE

The screenshot shows the SQL Developer interface with a query window containing four commands:

```
10 • CREATE TABLE Account (  
11     Account_No INT PRIMARY KEY,  
12     Customer_ID INT,  
13     Account_Type VARCHAR(20),  
14     Balance DECIMAL(12,2) DEFAULT 0,  
15     Branch VARCHAR(50),  
16     FOREIGN KEY (Customer_ID)  
17     REFERENCES Customers(Customer_ID)  
18 );
```

The Output window displays the execution results:

#	Time	Action	Message
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	10:17:58	USE BankDB	0 row(s) affected
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
✓ 4	10:21:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected

4. CREATE TRANSACTION TABLE

Query 1 physician Joins SQL File 5* SQL File 6* x

Limit to 1000 rows

```
19 • CREATE TABLE Bank_Transaction (  
20     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,  
21     Account_No INT,  
22     Transaction_Type VARCHAR(20),  
23     Amount DECIMAL(12,2),  
24     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,  
25     FOREIGN KEY (Account_No)  
26     REFERENCES Account(Account_No)  
27 );  
28
```

Output

Action Output

#	Time	Action	Message
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	10:17:58	USE BankDB	0 row(s) affected
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
✓ 4	10:21:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
✓ 5	10:22:40	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected

5. CREATE LOAN TABLE

```
28 • CREATE TABLE Loan (  
29     Loan_ID INT PRIMARY KEY,  
30     Customer_ID INT,  
31     Loan_Type VARCHAR(30),  
32     Loan_Amount DECIMAL(12,2),  
33     Interest_Rate DECIMAL(5,2),  
34     FOREIGN KEY (Customer_ID)  
35     REFERENCES Customers(Customer_ID)  
36 );  
37
```

Output

Action Output

#	Time	Action	Message
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	10:17:58	USE BankDB	0 row(s) affected
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
✓ 4	10:21:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
✓ 5	10:22:40	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected
✓ 6	10:23:52	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Lo...	0 row(s) affected

6. INSERT CUSTOMER DATA

```
37 • INSERT INTO Customers
38 (Customer_ID, Customer_Name, Phone, Email, City)
39 VALUES
40 (101, 'Rohit Sai', '9876543200', 'rohitsai@gmail.com', 'Hyderabad'),
41 (102, 'Adhvith', '9876543211', 'adhvith@gmail.com', 'Vijayawada'),
42 (103, 'Kiran', '9876543212', 'kiran@gmail.com', 'Bangalore'),
43 (104, 'Anirudh', '9876543213', 'anirudh@gmail.com', 'Chennai');
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected	
✓ 2	10:17:58	USE BankDB	0 row(s) affected	
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected	
✓ 4	10:21:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected	
✓ 5	10:22:40	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected	
✓ 6	10:23:52	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Lo...	0 row(s) affected	
✓ 7	10:26:37	INSERT INTO Customers (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Rohit Sai', '9876543...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	

7. INSERT ACCOUNT DATA

```
Query 1 physician Joins SQL File 5* SQL File 6* x
Limit to 1000 rows
44 • INSERT INTO Account
45 (Account_No, Customer_ID, Account_Type, Balance, Branch)
46 VALUES
47 (10001, 101, 'Savings', 50000, 'Hyderabad'),
48 (10002, 102, 'Savings', 75000, 'Vijayawada'),
49 (10003, 103, 'Current', 120000, 'Bangalore'),
50 (10004, 104, 'Savings', 45000, 'Chennai');
51
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected	
✓ 2	10:17:58	USE BankDB	0 row(s) affected	
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected	
✓ 4	10:21:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected	
✓ 5	10:22:40	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected	
✓ 6	10:23:52	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Lo...	0 row(s) affected	
✓ 7	10:26:37	INSERT INTO Customers (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Rohit Sai', '9876543...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	
✓ 8	10:28:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sav...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	

8. INSERT TRANSACTION DATA

```
51 • INSERT INTO Bank_Transaction
52 (Account_No, Transaction_Type, Amount)
53 VALUES
54 (10001, 'DEPOSIT', 10000),
55 (10002, 'DEPOSIT', 15000),
56 (10003, 'WITHDRAW', 20000),
57 (10004, 'DEPOSIT', 5000);
```

Output				
Action Output				
#	Time	Action	Message	
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected	
✓ 2	10:17:58	USE BankDB	0 row(s) affected	
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected	
✓ 4	10:21:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected	
✓ 5	10:22:40	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected	
✓ 6	10:23:52	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Lo...	0 row(s) affected	
✓ 7	10:26:37	INSERT INTO Customers (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Rohit Sai', '9876543...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	
✓ 8	10:28:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sav...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	
✓ 9	10:30:34	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0	

9. INSERT LOAN DATA

Query 1 physician Joins SQL File 5* SQL File 6* x

Limit to 1000 rows

```
58 • INSERT INTO Loan
59 (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
60 VALUES
61 (501, 101, 'Home Loan', 500000, 7.5),
62 (502, 102, 'Education Loan', 100000, 6.5),
63 (503, 103, 'Car Loan', 80000, 8.2),
64 (504, 104, 'Personal Loan', 50000, 10.5);
```

Output

Action Output

#	Time	Action	Message
✓ 1	10:17:56	CREATE DATABASE BankDB	1 row(s) affected
✓ 2	10:17:58	USE BankDB	0 row(s) affected
✓ 3	10:20:04	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NUL...	0 row(s) affected
✓ 4	10:21:05	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHA...	0 row(s) affected
✓ 5	10:22:40	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_N...	0 row(s) affected
✓ 6	10:23:52	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Lo...	0 row(s) affected
✓ 7	10:26:37	INSERT INTO Customers (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Rohit Sai', '9876543...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 8	10:28:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Sav...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 9	10:30:34	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000)...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 10	10:31:47	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'Hom...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0

10. DISPLAY TABLE DATA

Query 1 physician Joins SQL File 5* SQL File 6* x

Limit to 1000 rows

```
65 • SELECT * FROM Customers;
66
67 • SELECT * FROM Account;
68
69 • SELECT * FROM Bank_Transaction;
```

Result Grid

	Customer_ID	Customer_Name	Phone	Email	City
▶	101	Rohit Sai	9876543200	rohitsai@gmail.com	Hyderabad
	102	Adhvith	9876543211	adhvith@gmail.com	Vijayawada
	103	Kiran	9876543212	kiran@gmail.com	Bangalore
	104	Anirudh	9876543213	anirudh@gmail.com	Chennai
*	NULL	NULL	NULL	NULL	NULL

```
67 • SELECT * FROM Account;
68
69 • SELECT * FROM Bank_Transaction;
```

Result Grid

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
*	NULL	NULL	NULL	NULL	NULL

69 • `SELECT * FROM Bank_Transaction;`

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-16 10:30:34
2	10002	DEPOSIT	15000.00	2026-08-16 10:30:34
3	10003	WITHDRAW	20000.00	2026-08-16 10:30:34
4	10004	DEPOSIT	5000.00	2026-08-16 10:30:34
* NULL	NULL	NULL	NULL	NULL

71 • `SELECT * FROM Loan;`

72

73

74

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
* NULL	NULL	NULL	NULL	NULL

11. PROCEDURE TO FIND CUSTOMER ACCOUNTS

DELIMITER //

```
CREATE PROCEDURE GetCustomerAccounts(  
    IN p_Customer_ID INT  
)  
  
BEGIN  
    SELECT  
        C.Customer_ID,  
        C.Customer_Name,  
        A.Account_No,  
        A.Account_Type,  
        A.Balance,  
        A.Branch  
    FROM Customers C  
    JOIN Account A  
        ON C.Customer_ID = A.Customer_ID  
    WHERE C.Customer_ID = p_Customer_ID;  
END //  
  
DELIMITER ;  
  
CALL GetCustomerAccounts(101);
```

Result Grid						
Filter Rows:		Export:		Wrap Cell Content:		
	Customer_ID	Customer_Name	Account_No	Account_Type	Balance	Branch
▶	101	Rohit Sai	10001	Savings	50000.00	Hyderabad

Result 6 ×						
Output						
Action Output						
#	Time	Action	Message			
✓ 14	10:34:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned			
✓ 15	10:36:00	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Custome...	0 row(s) affected			
✓ 16	10:36:02	CALL GetCustomerAccounts(101)	1 row(s) returned			
✓ 17	10:37:38	CALL GetCustomerAccounts(101)	1 row(s) returned			

12. PROCEDURE TO DEPOSIT MONEY

Query 1 physician Joins SQL File 5* SQL File 6* ×

Limit to 1000 rows

92 DELIMITER //

93 CREATE PROCEDURE DepositMoney(

94 IN p_Account_No INT,

95 IN p_Amount DECIMAL(12,2)

96)

97 BEGIN

98 UPDATE Account

99 SET Balance = Balance + p_Amount

100 WHERE Account_No = p_Account_No;

101 END //

102 DELIMITER ;

103 CALL DepositMoney(10001, 5000);

Output

Action Output

#	Time	Action	Message
✓ 6	10:23:52	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), ...	0 row(s) affected
✓ 7	10:26:37	INSERT INTO Customers (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Rohit Sai', '98765...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 8	10:28:29	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, '...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 9	10:30:34	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 100...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 10	10:31:47	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'H...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
✓ 11	10:33:00	SELECT * FROM Customers LIMIT 0, 1000	4 row(s) returned
✓ 12	10:33:36	SELECT * FROM Account LIMIT 0, 1000	4 row(s) returned
✓ 13	10:34:05	SELECT * FROM Bank_Transaction LIMIT 0, 1000	4 row(s) returned
✓ 14	10:34:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
✓ 15	10:36:00	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Custome...	0 row(s) affected
✓ 16	10:36:02	CALL GetCustomerAccounts(101)	1 row(s) returned
✓ 17	10:37:38	CALL GetCustomerAccounts(101)	1 row(s) returned
✓ 18	10:39:01	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected
✓ 19	10:39:03	CALL DepositMoney(10001, 5000)	1 row(s) affected

13. PROCEDURE TO WITHDRAW MONEY

Query 1 physician Joins SQL File 5* SQL File 6* x

Limit to 1000 rows

```

104 DELIMITER //
105 CREATE PROCEDURE WithdrawMoney(
106     IN p_Account_No INT,
107     IN p_Amount DECIMAL(12,2)
108 )
109 BEGIN
110     UPDATE Account
111     SET Balance = Balance - p_Amount
112     WHERE Account_No = p_Account_No;
113 END //
114 DELIMITER ;
115 CALL WithdrawMoney(10001, 3000);

```

Output

Action Output

#	Time	Action	Message
9	10:30:34	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 100...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
10	10:31:47	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) VALUES (501, 101, 'H...	4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0
11	10:33:00	SELECT * FROM Customers LIMIT 0, 1000	4 row(s) returned
12	10:33:36	SELECT * FROM Account LIMIT 0, 1000	4 row(s) returned
13	10:34:05	SELECT * FROM Bank_Transaction LIMIT 0, 1000	4 row(s) returned
14	10:34:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
15	10:36:00	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Custome...	0 row(s) affected
16	10:36:02	CALL GetCustomerAccounts(101)	1 row(s) returned
17	10:37:38	CALL GetCustomerAccounts(101)	1 row(s) returned
18	10:39:01	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected
19	10:39:03	CALL DepositMoney(10001, 5000)	1 row(s) affected
20	10:40:23	CALL WithdrawMoney(10001, 3000)	Error Code: 1305. PROCEDURE bankdb.WithdrawMoney does not exist
21	10:40:30	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected
22	10:40:32	CALL WithdrawMoney(10001, 3000)	1 row(s) affected

14. TRIGGER: PREVENT NEGATIVE DEPOSIT

Query 1 physician Joins SQL File 5* SQL File 6* x

Limit to 1000 rows

```

119 DELIMITER //
120 CREATE TRIGGER CheckTransactionAmount
121 BEFORE INSERT ON Bank_Transaction
122 FOR EACH ROW
123 BEGIN
124     IF NEW.Amount <= 0 THEN
125         SIGNAL SQLSTATE '45000'
126         SET MESSAGE_TEXT =
127             'Transaction amount must be greater than zero';
128     END IF;
129 END //
130 DELIMITER ;

```

Output

Action Output

#	Time	Action	Message
12	10:33:36	SELECT * FROM Account LIMIT 0, 1000	4 row(s) returned
13	10:34:05	SELECT * FROM Bank_Transaction LIMIT 0, 1000	4 row(s) returned
14	10:34:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
15	10:36:00	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Custome...	0 row(s) affected
16	10:36:02	CALL GetCustomerAccounts(101)	1 row(s) returned
17	10:37:38	CALL GetCustomerAccounts(101)	1 row(s) returned
18	10:39:01	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected
19	10:39:03	CALL DepositMoney(10001, 5000)	1 row(s) affected
20	10:40:23	CALL WithdrawMoney(10001, 3000)	Error Code: 1305. PROCEDURE bankdb.WithdrawMoney does not exist
21	10:40:30	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected
22	10:40:32	CALL WithdrawMoney(10001, 3000)	1 row(s) affected
23	10:41:41	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	0 row(s) affected
24	10:41:43	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	Error Code: 1359. Trigger already exists
25	10:41:46	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	Error Code: 1359. Trigger already exists

15. CREATE TRANSACTION AUDIT TABLE

```
131 CREATE TABLE Transaction_Audit (  
132     Audit_ID INT PRIMARY KEY AUTO_INCREMENT,  
133     Transaction_ID INT,  
134     Account_No INT,  
135     Transaction_Type VARCHAR(20),  
136     Amount DECIMAL(12,2),  
137     Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP  
138 );
```

Output:

Action Output

#	Time	Action	Message
✓ 13	10:34:05	SELECT * FROM Bank_Transaction LIMIT 0, 1000	4 row(s) returned
✓ 14	10:34:19	SELECT * FROM Loan LIMIT 0, 1000	4 row(s) returned
✓ 15	10:36:00	CREATE PROCEDURE GetCustomerAccounts(IN p_Customer_ID INT) BEGIN SELECT C.Custome...	0 row(s) affected
✓ 16	10:36:02	CALL GetCustomerAccounts(101)	1 row(s) returned
✓ 17	10:37:38	CALL GetCustomerAccounts(101)	1 row(s) returned
✓ 18	10:39:01	CREATE PROCEDURE DepositMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected
✓ 19	10:39:03	CALL DepositMoney(10001, 5000)	1 row(s) affected
✗ 20	10:40:23	CALL WithdrawMoney(10001, 3000)	Error Code: 1305. PROCEDURE bankdb.WithdrawMoney does not exist
✓ 21	10:40:30	CREATE PROCEDURE WithdrawMoney(IN p_Account_No INT, IN p_Amount DECIMAL(12,2)) BEGIN ...	0 row(s) affected
✓ 22	10:40:32	CALL WithdrawMoney(10001, 3000)	1 row(s) affected
✓ 23	10:41:41	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	0 row(s) affected
✗ 24	10:41:43	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	Error Code: 1359. Trigger already exists
✗ 25	10:41:46	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	Error Code: 1359. Trigger already exists
✓ 26	10:44:22	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID ...	0 row(s) affected

16. TRIGGER: TRANSACTION AUDIT

```
Query 1 physician Joins SQL File 5* SQL File 6*  
139 DELIMITER //  
140 CREATE TRIGGER TransactionAudit  
141 AFTER INSERT ON Bank_Transaction  
142 FOR EACH ROW  
143 BEGIN  
144     INSERT INTO Transaction_Audit  
145     (  
146         Transaction_ID,  
147         Account_No,  
148         Transaction_Type,  
149         Amount  
150     )  
151     VALUES  
152     (  
153         NEW.Transaction_ID,  
154         NEW.Account_No,  
155         NEW.Transaction_Type,  
156         NEW.Amount  
157     );  
158 END //  
159 DELIMITER ;
```

Output:

Action Output

#	Time	Action	Message
✗ 24	10:41:43	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	Error Code: 1359. Trigger already exists
✗ 25	10:41:46	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	Error Code: 1359. Trigger already exists
✓ 26	10:44:22	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID ...	0 row(s) affected
✓ 27	10:45:30	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INS...	0 row(s) affected


```

160 • INSERT INTO Bank_Transaction
161     (Account_No, Transaction_Type, Amount)
162     VALUES
163     (10001, 'DEPOSIT', 2500);

```

Output

Action Output

#	Time	Action	Message
✖ 25	10:41:46	CREATE TRIGGER CheckTransactionAmount BEFORE INSERT ON Bank_Transaction FOR EACH ROW BE...	Error Code: 1359. Trigger already exists
✔ 26	10:44:22	CREATE TABLE Transaction_Audit (Audit_ID INT PRIMARY KEY AUTO_INCREMENT, Transaction_ID ...	0 row(s) affected
✔ 27	10:45:30	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INS...	0 row(s) affected
✖ 28	10:45:32	CREATE TRIGGER TransactionAudit AFTER INSERT ON Bank_Transaction FOR EACH ROW BEGIN INS...	Error Code: 1359. Trigger already exists
✔ 29	10:46:32	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 2500)	1 row(s) affected

17. PROCEDURE: ACCOUNT TRANSFER

```

167     IN ReceiverAccount INT,
168     IN TransferAmount DECIMAL(12,2)
169 )
170 BEGIN
171     DECLARE SenderBalance DECIMAL(12,2);
172     SELECT Balance
173     INTO SenderBalance
174     FROM Account
175     WHERE Account_No = SenderAccount;
176     IF SenderBalance < TransferAmount THEN
177         SIGNAL SQLSTATE '45000'
178         SET MESSAGE_TEXT =
179         'Transfer failed: Insufficient balance';
180     ELSE
181         UPDATE Account
182         SET Balance = Balance - TransferAmount
183         WHERE Account_No = SenderAccount;
184         UPDATE Account
185         SET Balance = Balance + TransferAmount
186         WHERE Account_No = ReceiverAccount;
187     END IF;
188 END //
189 DELIMITER ;

```

Output

Action Output

#	Time	Action	Message
✖ 30	10:49:25	CALL TransferMoney(10001, 10002, 5000)	Error Code: 1305. PROCEDURE bankdb.TransferMoney does not exist
✔ 31	10:50:31	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount INT, IN Transfer...	0 row(s) affected
✔ 32	10:50:34	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected

```

191 • SELECT *
192 FROM Account
193 WHERE Account_No IN (10001, 10002);

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: [fA](#)

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	47000.00	Hyderabad
10002	102	Savings	80000.00	Vijayawada

Account 7 x

Output

Action Output

#	Time	Action	Message
31	10:50:31	CREATE PROCEDURE TransferMoney(IN SenderAccount INT, IN ReceiverAccount INT, IN Transfer...	0 row(s) affected
32	10:50:34	CALL TransferMoney(10001, 10002, 5000)	1 row(s) affected
33	10:51:08	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned

18. PROCEDURE WITH IN, OUT PARAMETERS

Query 1 physician Joins SQL File 5* SQL File 6* x

Limit to 1000 rows

```

194 DELIMITER //
195 • CREATE PROCEDURE GetBalance(
196     IN p_Account_No INT,
197     OUT p_Balance DECIMAL(12,2)
198 )
199 BEGIN
200     SELECT Balance
201     INTO p_Balance
202     FROM Account
203     WHERE Account_No = p_Account_No;
204
205 END //
206 DELIMITER ;
207 • CALL GetBalance(10001, @CurrentBalance);
208 • SELECT @CurrentBalance;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [fA](#)

@CurrentBalance
47000.00

Result 8 x

Output

Action Output

#	Time	Action	Message
33	10:51:08	SELECT * FROM Account WHERE Account_No IN (10001, 10002) LIMIT 0, 1000	2 row(s) returned
34	10:53:17	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMAL(12,2)) BEGIN ...	0 row(s) affected
35	10:53:20	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned

19. TRIGGER COMMANDS

```
209 • SHOW TRIGGERS;
210
211
212
213
214
215
---
```

Result Grid

Trigger	Event	Table	Statement	Timing	Created	sql_mode	Definer
CheckTransactionAmount	INSERT	bank_transaction	BEGIN IF NEW.Amount <= 0 THEN SIGN...	BEFORE	2026-08-16 10:41:41.10	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	root
TransactionAudit	INSERT	bank_transaction	BEGIN INSERT INTO Transaction_Audit (... AFTER	AFTER	2026-08-16 10:45:30.33	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	root

Result 9

Output

Action Output

#	Time	Action	Message
34	10:53:17	CREATE PROCEDURE GetBalance(IN p_Account_No INT, OUT p_Balance DECIMAL(12,2)) BEGIN ...	0 row(s) affected
35	10:53:20	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned
36	10:53:57	SHOW TRIGGERS	2 row(s) returned

20. PROCEDURE MANAGEMENT COMMANDS

```
210 • SHOW PROCEDURE STATUS
211 WHERE Db = 'BankDB';
212
213
214
215
---
```

Result Grid

Db	Name	Type	Definer	Modified	Created	Security_type	Comment	character_set_client	collation_connection	Datal Colla
bankdb	DepositMoney	PROCEDURE	root@localhost	2026-08-16 10:39:01	2026-08-16 10:39:01	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8m
bankdb	GetBalance	PROCEDURE	root@localhost	2026-08-16 10:53:17	2026-08-16 10:53:17	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8m
bankdb	GetCustomerAccounts	PROCEDURE	root@localhost	2026-08-16 10:36:00	2026-08-16 10:36:00	DEFINER		utf8mb4	utf8mb4_0900_ai_ci	utf8m
bankdb	TransferMoney	PROCEDURE	root@localhost	2026-08-16 10:50:31	2026-08-16 10:50:31	DEFINER		utf8mb4	utf8mb4_0900 ai ci	utf8m

Result 10

Output

Action Output

#	Time	Action	Message
35	10:53:20	SELECT @CurrentBalance LIMIT 0, 1000	1 row(s) returned
36	10:53:57	SHOW TRIGGERS	2 row(s) returned
37	10:54:51	SHOW PROCEDURE STATUS WHERE Db = 'BankDB'	5 row(s) returned